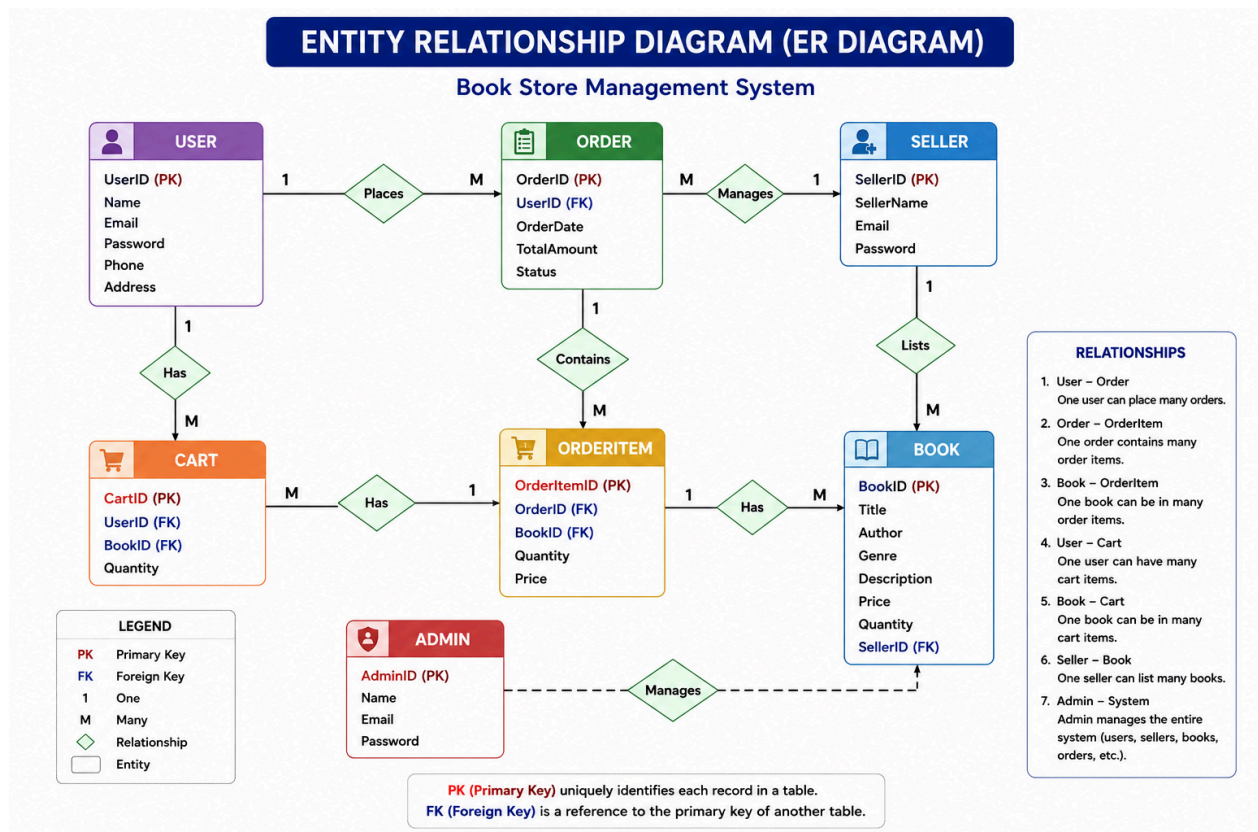


ENTITY RELATION DIAGRAM (ER DIAGRAM)

The **Entity Relationship Diagram (ER Diagram)** represents the database structure of the Book Store Management System and shows the relationships between the main entities involved in the application.

The main entities in the Book Store project are:

- User
- Book
- Cart
- Order
- OrderItem
- Seller
- Admin



1. User–Book Relationship

Type: Many-to-Many (M:M)

A single user can browse, search, review, or purchase many books, and a single book can be viewed or purchased by many users.

Implementation:

This relationship can be represented indirectly through **Cart**, **OrderItem**, or an optional **Interaction** entity that stores user-book related activities such as reviews, ratings, or browsing history.

2. User–Order Relationship

Type: One-to-Many (1:M)

A single user can place multiple orders, but each order belongs to one user.

Implementation:

The **Order** entity contains **UserID** as a foreign key/reference to track which user placed the order.

3. Book–OrderItem Relationship

Type: One-to-Many (1:M)

A single book can appear in many order items, but each order item refers to one specific book.

Implementation:

The **OrderItem** entity contains **BookID** as a foreign key/reference.

4. Order–OrderItem Relationship

Type: One-to-Many (1:M)

A single order can contain multiple order items, but each order item belongs to one order only.

Implementation:

The **OrderItem** entity contains **OrderID** as a foreign key/reference.

5. User–Cart Relationship

Type: One-to-Many (1:M)

A single user can have multiple cart items, but each cart item belongs to one user.

Implementation:

The **Cart** entity contains **UserID** as a reference to identify the owner of the cart item.

6. Book–Cart Relationship

Type: One-to-Many (1:M)

A single book can be added to multiple carts, but each cart entry refers to one book.

Implementation:

The **Cart** entity contains **BookID** as a foreign key/reference.

7. Seller–Book Relationship

Type: One-to-Many (1:M)

A single seller can list multiple books, but each book belongs to one seller.

Implementation:

The **Book** entity can contain **SellerID** to indicate which seller listed the book.

8. Admin–System Relationship

Type: One-to-Many / Management Relationship

The admin manages users, sellers, books, orders, and the overall platform activities.

Implementation:

Admin does not always require a direct database relationship like other entities, but admin privileges are used to control and manage records in the system.

Main Entities and Attributes

User

- UserID
- Name
- Email
- Password
- Phone
- Address

Book

- BookID

- Title
- Author
- Genre
- Description
- Price
- Quantity
- SellerID

Cart

- CartID
- UserID
- BookID
- Quantity

Order

- OrderID
- UserID
- OrderDate
- TotalAmount
- Status

OrderItem

- OrderItemID
- OrderID
- BookID
- Quantity
- Price

Seller

- SellerID
- SellerName
- Email
- Password

Admin

- AdminID
- Name
- Email
- Password

